

**Artificially intelligent cities:
Algorithmic control negotiated by human unreason**

by

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Dedication

This dissertation is directed toward the people living in urban areas undergoing the transformation of various “Smart City” initiatives. It is meant for everyone who believes in the political importance of free public space. It is written for those affected by the increasing power of planetary computation—which is to say, almost all of us. Most importantly, it is dedicated to the people in charge of building these very cities and initiating smart city proposals: the politicians, urbanists, architects, developers, technologists and other urban actors who carry the responsibility for the citizens of their playgrounds.

A Note on the Text

In this dissertation, I use some terms from architecture, technology, and philosophy that might be unfamiliar or have a very specific meaning in this context. Explaining those words directly in the text would decrease the flow and readability. I have put together a **Glossary** at the end of this document. This glossary does sometimes contain direct definitions of these terms but more often explain what they mean in the context that this dissertation is situated in. Feel free to refer to the glossary whenever you come across a word that feels confusing or if you want to double-check its meaning in this text. Reading the chapters alongside this glossary will help make the concepts clearer.

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Abstract

Every day the built environment becomes more monitored, more automated, more responsive, and more controlled. Digital technologies are diffusing into cities, streets, and homes. Networked devices colonize urban spaces. They sense, reason, and act, turning the city into an automated system. Today, this effort is largely done by private institutional actors under the pretense of “smart” or “cognitive” cities.

Institutional “smart” cities weaponize urban technology. Sensing becomes hidden surveillance, reasoning becomes algorithmic prediction, and acting becomes spatial control. They reduce the human to a behavioral profile—constantly monitored, preempted, and manipulated. The resulting condition contradicts the city as a stage for liberty, equality, and fraternity—values now replaced by the technocratic ideals of comfort, security, and sustainability. It eliminates the unpredictability of complex human nature that defines public life and constructs a frictionless and automated experience.

The dissertation investigates the power of urban networks and demands an ideological reversal. It accepts the inevitable intrusion of technology and uses it as a support for spatial negotiation guided by three core principles: participation, transparency, and friction. The dissertation introduces the “reversed cyborg”, an augmentation of computational infrastructures by human peculiarity; dreams, unreason, and desires.

The resulting framework presents an alternative for future cities. It preserves the citizen as the protagonist of the urban narrative, resisting a reduction to a data source. It uses urban technology to support collective negotiation as a core necessity of democratic cities. The goal is to ensure that shared environments remain fundamentally human: contested, unpredictable, and alive.

Introduction : The Diffusion of the Screen

Imagine it's Wednesday afternoon, 2:45 pm, 25th of February 2026—well, at least that's what the Samsung watch tells you. You are sitting in a café in Paris. It was recommended by Google's algorithm. The table you are sitting at was cut by a CNC machine. You scan the QR code on the table. It connects you directly to the digital menu to let you place your order. Nice, they actually have this new salty coffee you repeatedly saw on Instagram. That's why you came here. Meta's algorithms lured your desire for consumption by showing you promotional videos. You select the coffee and place the order. Just a few minutes later, the waiter brings the salty coffee. You pay by tapping your phone against the register. Your watch beeps; it read your calendar: 3 pm, time to join the lecture. Microsoft's algorithm suggested this lecture through your LinkedIn network. You open your new laptop. A YouTube video convinced you to buy it from Amazon. The device made a long journey: from extraction mines to automated manufacturing facilities, to Amazon's logistics network, and finally to your doorstep...

The Wednesday Afternoon scenario illustrates how two domains are merging. The physical and digital spaces are not distinct anymore; they are increasingly inseparable. The QR code, the register, the online lecture, and the package delivery are all examples of an interconnected cyber-physical world.

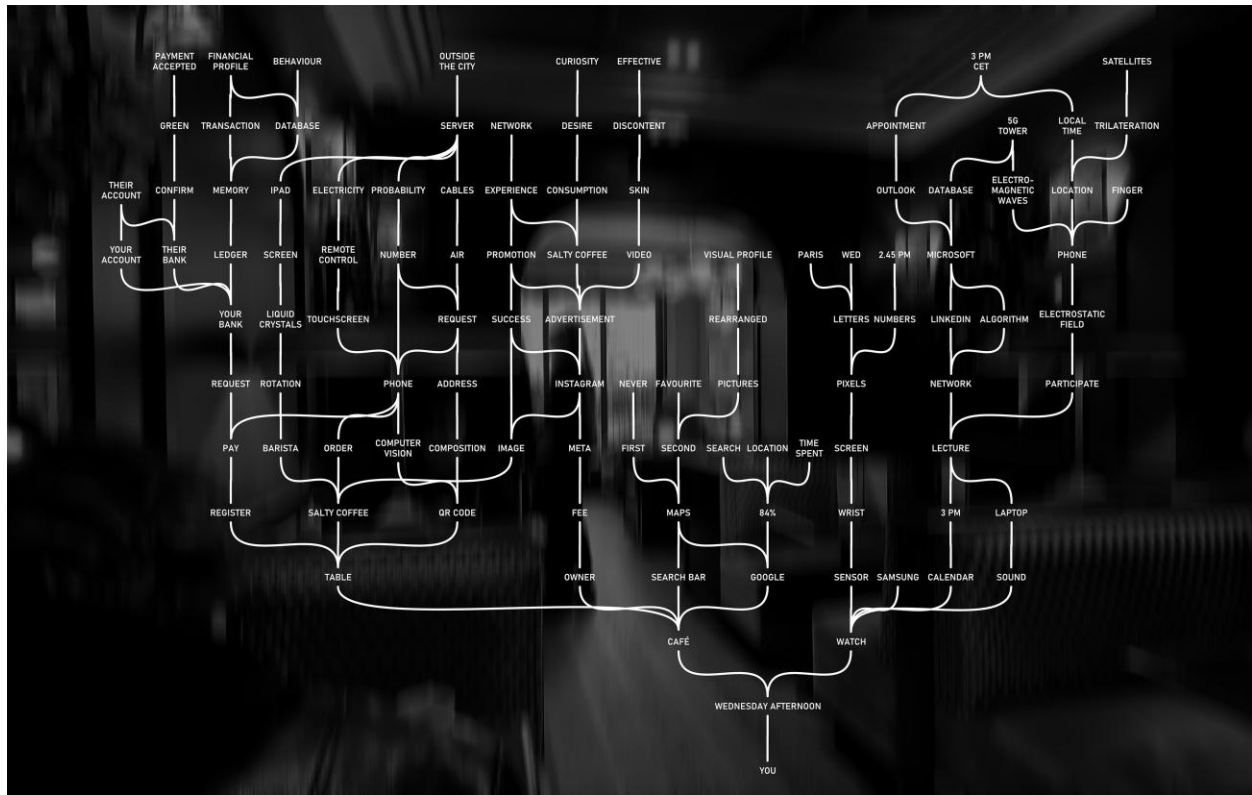


Image 0: Correlation Tree of significant things on an average Wednesday Afternoon

For decades, human interaction with the digital world happened almost exclusively through the familiar, glowing rectangles: phones, computers, television, and other screens. French theorist Paul Virilio observed long before the arrival of the smartphone that the screen was destined to open out and replace not just the window, "but even the 'doorway' itself,"¹ fundamentally changing the role of the house or apartment. Virilio warned of a condition he called "polar inertia,"¹ a state where physical human movement is increasingly replaced by the arrival of information at the speed of light and the "electro-magnetic telekinesis,"¹ the ability to remotely control physical surroundings through telecommunication networks. In this new reality, people do not simply inhabit buildings; instead, electronic systems actively surround them, shaping their environment and daily habits.

¹ Virilio, P. (1990). Polar Inertia. SAGE Publications.

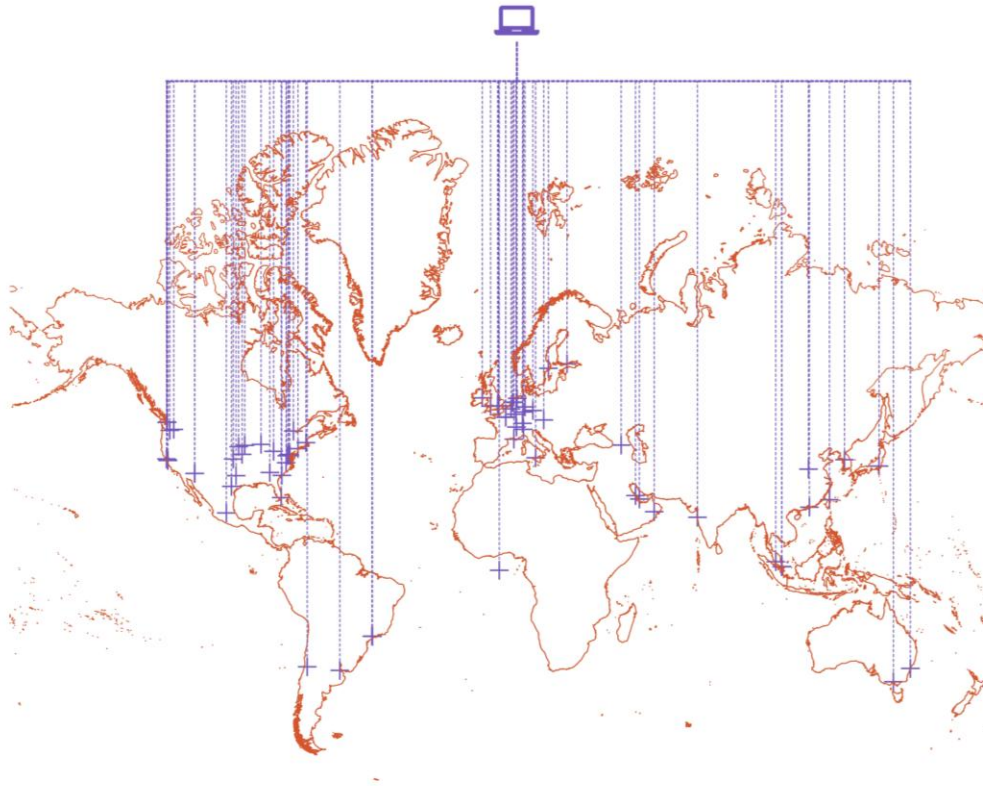


Image 2: Communication to server locations from my laptop in Paris during 3 hours of doing research

As the design theorist Benjamin Bratton defines it, an interface is "any point of contact between two complex systems that governs the conditions of exchange between those systems"². The digital screen used to be the primary interface, acting as a clear boundary between the physical world and the digital realm. Today, however, the technologies that control those screens are diffusing into the physical environment of contemporary cities. The urban landscape is no longer a passive scaffold for human activity; it is being transformed into a cyber-physical system with the autonomous ability to **sense, reason, and act**. This turns the built fabric into an interface, an active mediator between its inhabitants and the disembodied computational systems that control it.

² Bratton, B. H. (2015). *The Stack: On Software and Sovereignty*. MIT Press.

This infiltration of the built environment leads to what can be described as the death of the traditional interface. While physical screens certainly still exist, their fundamental model of interaction is being replaced. As long as the main interface was a screen, the interaction went both ways, requiring conscious, voluntary human input. Emerging systems are removing the need for human action altogether. They model and predict human behavior to preemptively control the environment. As Shoshana Zuboff warns, the goal of these systems is no longer simply to automate information flows, but to "automate us"³. When the environment perfectly anticipates human behavior and acts before an individual makes a choice, the traditional two-way exchange dies. People are increasingly stopping looking at interfaces, because the system is already looking at them.

This dissertation investigates this contemporary phenomenon and its implications for public urban life. First, Chapter 1 will analyze the mechanics of cyber-physical systems to understand how the city autonomously senses, reasons, and acts. Second, Chapter 2 will investigate the status quo, showing how private institutional actors drive this transition in the name of surveillance, prediction, and control, effectively reducing the human to a passive data profile. Chapter 3 establishes a manifesto proposing three alternative core principles: participation, transparency, and friction. The proposal accepts the inevitable intrusion of new technologies, but preserves people as the protagonists of the urban narrative.

³ Zuboff, S. (2019). *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power*. PublicAffairs.

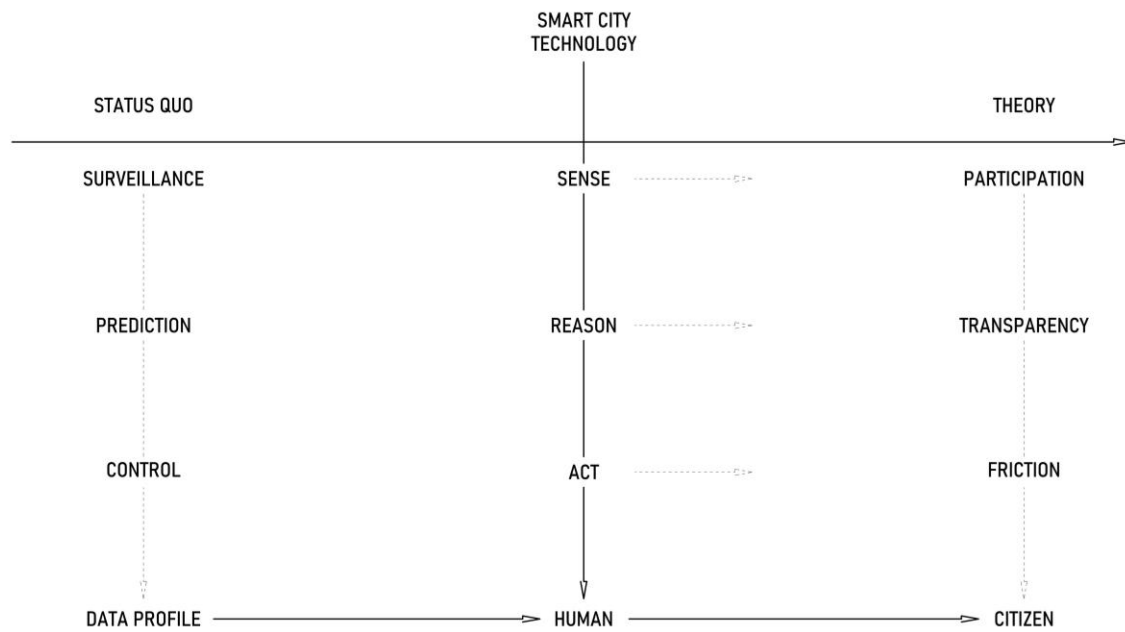


Image 3: Diagram to organize the key chapters of the argumentation into a grid of intentionality.

Chapter 1 : The Feedback Loop

This new urban reality is based on a continuous, automated feedback loop rather than a specific architectural style or single technology. The city is becoming a global algorithmic system that operates in three sequential phases: sensing, reasoning, and acting.

Examining this three-part process reveals how the built environment has changed from a static fabric into a dynamic machine capable of interacting with its inhabitants in real-time. According to recent spatial planning studies, an estimated 150 billion new networked devices and sensors are expected to be installed within the current decade alone.⁴ While each device controls a tiny fraction of the environment, their networked whole creates a new operational reality that is analyzed in this chapter.

1.1 Sense

The first phase of the feedback loop is the collection of data from the physical world. To interact with the city, the algorithmic system must first be able to perceive it. However, a machine does not feel or experience the world; it measures it. This measurement is achieved through a massive deployment of sensors embedded directly into the urban environment.

Rapidly increasing, cameras, microphones, infrared detectors, LiDAR, thermal sensors, radar, and pressure pads are woven into urban streets and buildings. These technologies allow the system to perceive the environment in ways that far exceed human capabilities. For instance, millimeter-wave radar can penetrate non-conductive materials to monitor micro-displacements of the chest, effectively tracking human vital signs like breathing and heart rates from a distance. A

⁴ A. Grêt-Regamey & M. Switalski, "Digital Twin for Urban Planning in the Green Deal Era" (2021).

possible scenario is a sensor built into a wall across the street that measures a sudden increase in your heart rate when you receive a call from your boss.

These devices are often added to lampposts, built into sidewalks, or hidden in walls. They quietly register the presence and movement of pedestrians, vehicles, and weather patterns, monitoring everything from air quality and ambient temperature to the specific gait, heart rate, and micro-expressions of the people passing by. The theorist Adam Greenfield highlights the impact of this phenomenon in *Radical Technologies*, describing the contemporary streetscape as "an urban fabric furiously siphoning up information, every square meter of seemingly banal sidewalk yielding so much data about its uses and its users that nobody quite yet knows what to do with it all."⁵ And yet, they are almost imperceptible to the public. They often appear as small devices of no particular form.

The corporate goal behind this network is clearly stated by multinational technology vendor Siemens. They imagine a future relying on "countless autonomous, intelligently functioning IT systems that will have perfect knowledge of users' habits and energy consumption." The ultimate goal is to "optimally regulate and control resources by means of autonomous IT systems."⁵

Sensing is not neutral perception; it is a mechanism that turns the complex, analog reality of human cities into standardized numerical data. Through this translation, the physical world is digitized, creating a continuous stream of raw data. In a developed system, every moment of hesitation, every chosen route, and every physical interaction will be extracted and compiled into data profiles. This process ensures that the unpredictable, messy nature of human life is flattened into calculable metrics, ready for the next phase of the loop.

⁵ Greenfield, A. (2017). *Radical Technologies: The Design of Everyday Life*. Verso Books.

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Image 4: My Google Data: most recent search history

1.2 Reason

Once the environment has been sensed and quantified, this vast amount of data must be processed. This leads to the "reasoning" phase of the loop. This phase is about finding patterns, drawing connections, making calculations, and finally creating predictions. The processing does not happen within the sensor itself or on the street corner. Instead, the data is transmitted through optic fiber cables and wireless networks to large data centers, commonly referred to as the "Cloud".

Despite its name, the Cloud is a massive, heavy industrial machine. It relies on a planetary-scale infrastructure made of subterranean wires, fiber cables crossing the oceans, energy grids, and vast server farms that consume enormous amounts of water and electricity to maintain their continuous operations. This infrastructure represents a new planetary-scale condition, where computation and ecology are deeply bound⁶. And it is exactly this distributed machine that is given the task to understand what has happened, what is happening, and what will happen.

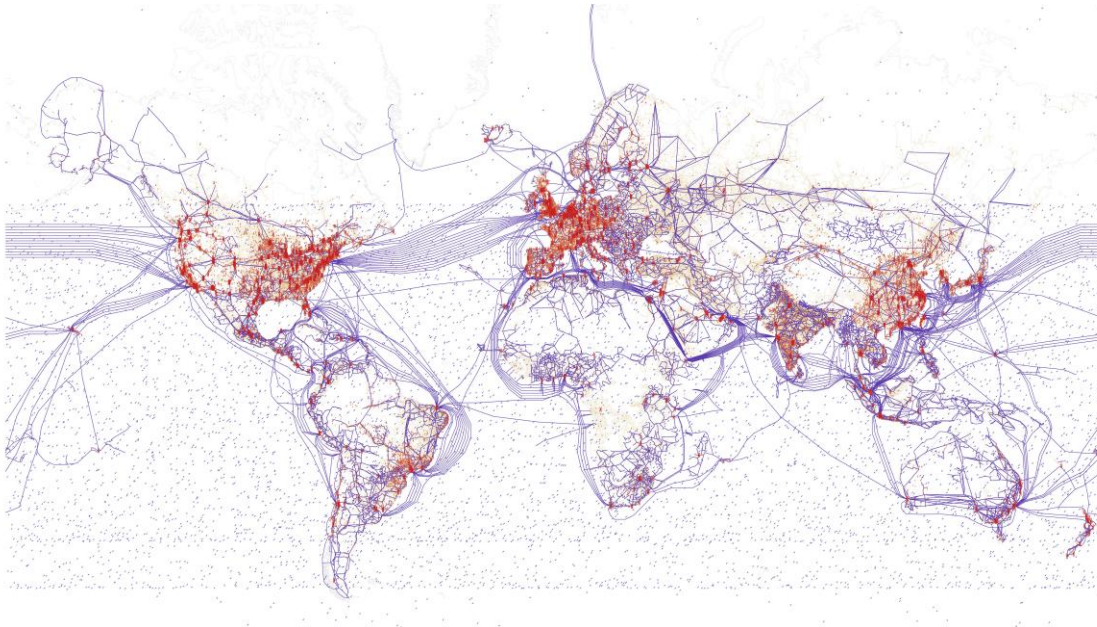


Image 5: Attempt to map the planetary computational apparatus. (red: datacenters, orange: power plants, yellow: electricity grid, purple lines: optic fiber cables, purple dots: Starlink satellites)

⁶ Gilman, N. (Ed.). (2024). *The Planetary*. Antikythera.

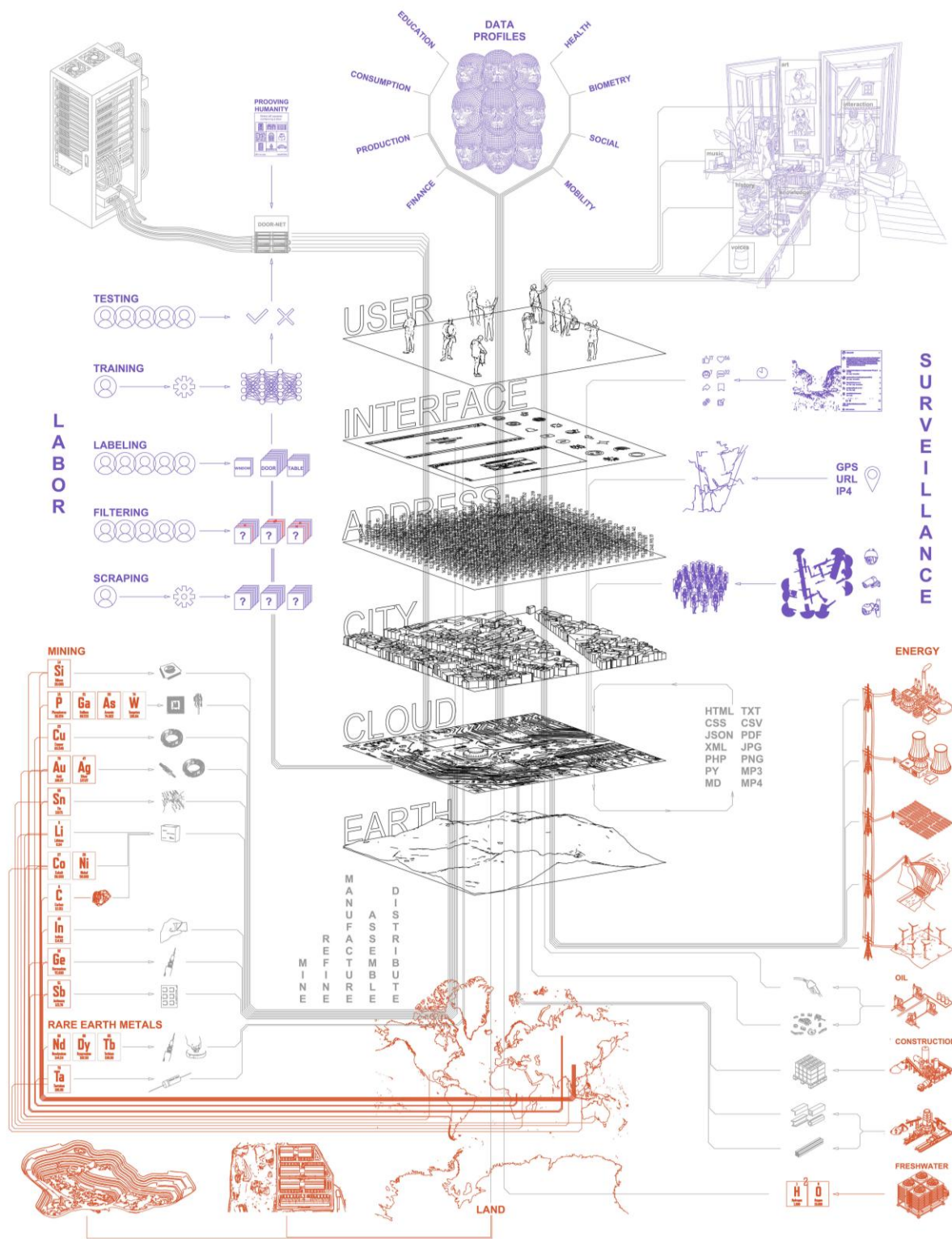


Image 6: Implications of the Computational Infrastructure both for the Earth and for the Human rooted in Benjamin Bratton's Stack

It is within this infrastructure that the "reasoning" takes place. However, one must be careful not to confuse algorithmic processing with human thought. As historian Yuval Noah Harari points out, human beings generally struggle to weigh massive amounts of data; human reason suffers from the "fallacy of the single cause," preferring to find one clear, conscious reason for why something happens. Algorithms, by contrast, excel at taking hundreds or thousands of seemingly unrelated data points and simultaneously weighing them against each other to find hidden correlations.⁷

This leads to the presumption that we just need to feed the machine more data until it eventually succeeds to solve any problem we couldn't solve yet and know anything we don't know yet. Harari calls this the "naive view" of information, David Brooks from the New York Times calls it "Data-ism". It is the believe that more information "informs" us about reality and that more data leads to more truth. Instead, Harari argues, that the true power of information is to "put us in formation"⁷. It relates bodies, clicks, and transactions to new social and political constructs. It connects distant behaviors and steers human pathways.

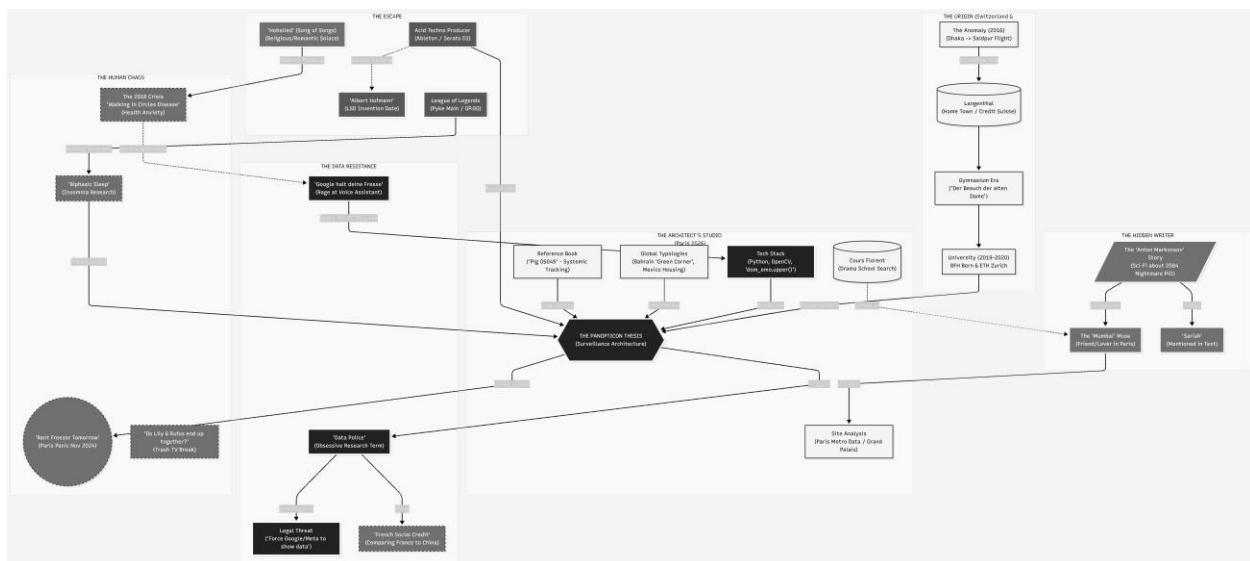


Image 7: Diagram of Connections that Gemini (Google's LLM) constructed based on my Google Search History

⁷ Yuval Noah Harari, Nexus (2024). (Extract on the "fallacy of the single cause").

To grasp the implications, it is important to understand how these modern algorithms work. Historically, algorithms were deductive. They operated on strict, pre-programmed rules: if a specific event happens, then execute a specific command. Today, the systems governing contemporary cities rely heavily on inductive machine learning. These algorithms look at massive datasets, identify historical patterns, and induce their own rules. They process the data to find patterns and make statistical bets on what will happen next. The system does not reason by understanding human meaning, but by mathematically correlating variables to predict an outcome before it occurs. And since the system is part of the feedback loop, it can continuously adapt its objectives based on the outcomes of its own activities.

As AI scholar Kate Crawford explains, this form of reasoning requires an "epistemological flattening."⁸ The messy complexity of human life and physical space is isolated from its context and flattened into a clean, computable signal. When the system "reasons," it is not trying to understand the meaning behind a citizen's action. It does not know why people gather in a square or why a person is running. It merely recognizes that a specific mathematical pattern of data is occurring and draws connections accordingly.

By comparing this real-time data against its vast archives of past events, the algorithm calculates probabilities. The final output of this reasoning phase is a prediction. At its core, every calculation is a statistical bet on what is most likely to happen next. The system uses these probabilities to anticipate the immediate future, transforming raw data into actionable instructions ready to be executed in the final phase of the loop.

1.3 Act

The third step of the feedback loop is actuation. If the system “only” sensed and reasoned, it would merely be an observation tool. The loop closes when the reasoning from the Cloud is translated back into physical changes in the local urban environment.

⁸ Crawford, K. (2021). *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*.

This is achieved through Cyber-Physical Systems (CPS): engineered networks that integrate digital algorithms with physical components. While sensors gather data, actuators are the mechanisms that alter the material world. They take many forms. Electric motors spin gears to transform static elements into moving structures. Electromagnets secure doors or allow architectural partitions to slide into new configurations. LED systems adjust color temperatures and ambient lighting. Valves emit gases to change the composition of the air. Through these parts, the city physically responds to the algorithmic commands it receives.

To understand the potential spatial impact of this actuation, the modern logistics warehouse serves as a prototype for the future city. In his analysis of Amazon fulfillment centers, Jesse LeCavalier describes how autonomous robotic drive units constantly move inventory pods based on algorithmic predictions of consumer demand. Once an item is picked, the robot brings the shelf not to its original position, but to the closest open slot. Through this continuous, automated movement, the physical architecture of the warehouse is constantly reconfiguring itself.⁹ Liam Young notes that spaces like these are fundamentally "machine landscapes." They are organized by strict efficiency algorithms and are becoming "human exclusion zones". They demonstrate a new architectural typology that does not relate to human patterns of orientation or habitation.¹⁰

Through these systems, the digital and physical spaces are tightly linked. The actuation happens automatically, adjusting and acting without requiring any conscious or active input from the human citizen. Once the system acts, the sensors immediately record the city's new state and the citizen's response. This fresh data streams back to the Cloud, and the loop begins again.

⁹ Jesse LeCavalier, "Human Exclusion Zones," in *Machine Landscapes: Architectures of the Post-Anthropocene*, ed. Liam Young (Architectural Design, 2019), 52.

¹⁰ Liam Young, "Neo-Machine: Architecture Without People," in *Machine Landscapes: Architectures of the Post-Anthropocene* (Architectural Design, 2019), 8.

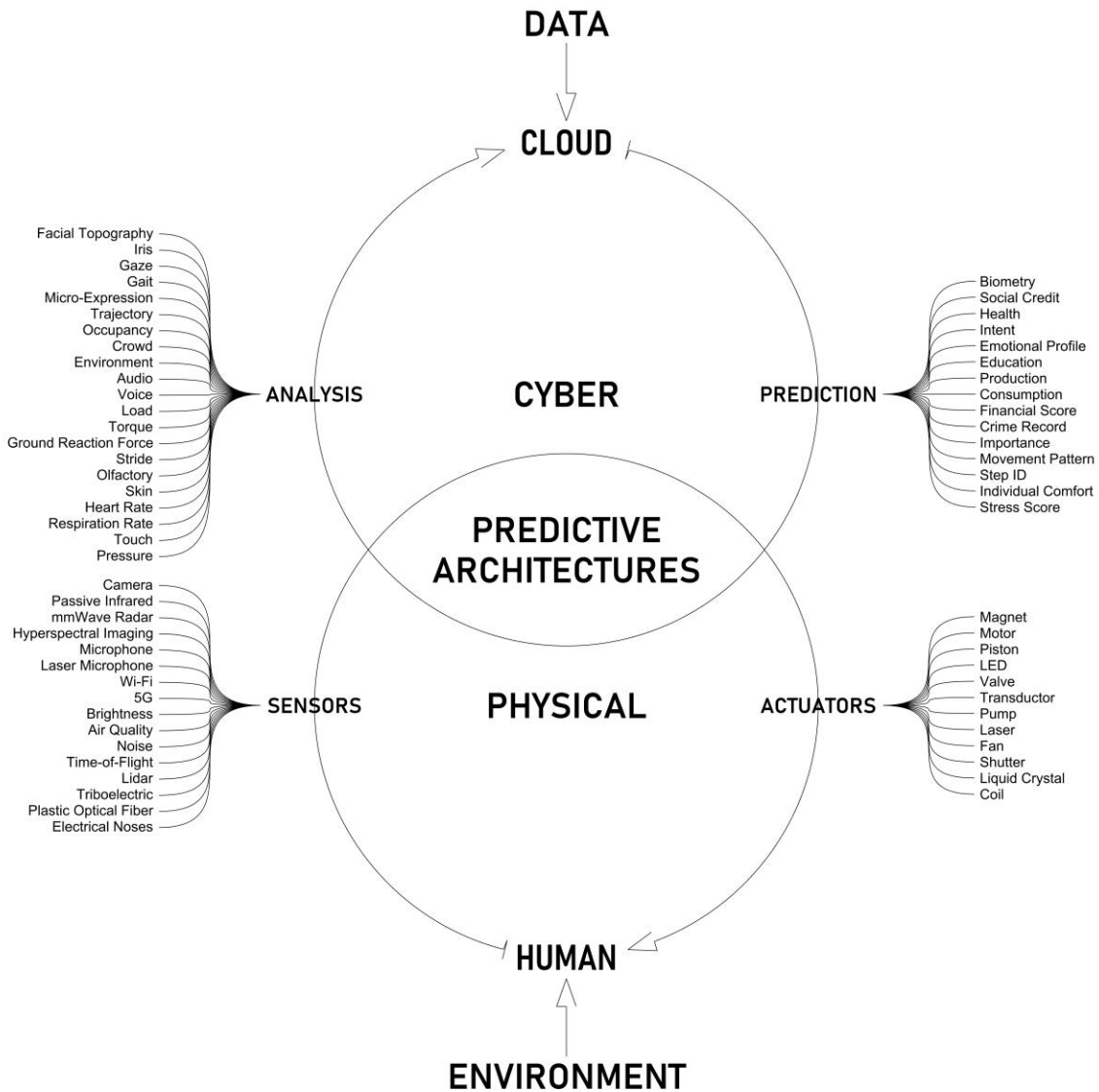


Image 8: The Predictive Feedback Loop: The chapters “sense” and “act” can directly be assigned to sensors and actuators. The chapter “reason” includes everything connected to the upper circle: analysis, cloud, data, and prediction.

Chapter 2 : The Status Quo of Smart Cities

Chapter 1 analyzed the feedback loop's functional anatomy: sensing, reasoning, and acting. This chapter shifts the focus from operation to operators. Who builds these systems, and why? The smart city's loop is not a neutral technological infrastructure, but a highly political spatial project¹¹. It materializes the incentives of the corporations and institutions that own it. Underneath every glossy poster of comfort, security, and sustainability run hidden agendas and deeper strategies.

For instance, Singapore exemplifies a state-level application of widespread monitoring. The GovTech agency is installing the "Lamp-post-as-a-Platform" project. This network deploys 100,000 surveillance cameras linked to facial recognition software across the city-state¹¹. The government retains control over the collected data. The earlier Computer Misuse and Cybersecurity Act can be used to justify reasoning behavioral signals. In South Korea, New Songdo City represents the smart city built from scratch. Cisco Systems wired much of the urban fabric with sensor networks. These sensors monitor traffic, waste, and energy flow, linking the city to corporate network infrastructures¹¹. Similarly, in Rio de Janeiro, the IBM Operations Center centralizes the management of traffic, emergencies, and utilities. It treats the city as a top-down optimization problem to be solved by IBM's prediction models¹¹.

In Shenzhen, algorithmic tools already support traffic enforcement and policing through camera networks and AI-based image analysis. China's Social Credit System is a state-led ledger of blacklists and redlists that aggregates judicial, administrative, and commercial data to rate the "trustworthiness" of individuals. People placed on blacklists can be blocked from purchasing tickets for flights and high-speed trains. Who is permitted passage and who is flagged, slowed, or excluded increasingly depends on database calculations.

¹¹ Figueiredo, S. M., Krishnamurthy, S., & Schroeder, T. (Eds.). (2019). *Architecture and the Smart City*. Routledge.

The following three sections interrogate the inherently political nature of this apparatus. The first examines how urban *sensing* is weaponized as **surveillance**, harvesting environmental data to extract human experience. The second analyzes how algorithmic *reasoning* becomes **prediction**, deploying planetary cloud infrastructures to preempt and automate choice. The third uncovers how spatial *acting* is weaponized as **control**, dynamically adjusting physical environments to create a frictionless terrain that eliminates democratic unpredictability. The feedback loop reduces the rich complexity of human citizens into numerical data profiles. Daily habits, bodily movements, and social interactions become calculable metrics. The citizen becomes both a data source and a puppet.

2.1 Surveillance

Urban sensing technologies are installed as tools for monitoring air, traffic, or energy. However, the same technologies that have been deployed for such reasons collect endless amounts of data about human behavior. This collection is no accidental byproduct. It is a systematic business model.

Sensor networks cover public space to manufacture what Shoshana Zuboff identifies as behavioral surplus—data beyond service improvement, fed into predictive markets¹². A streetlight senses your trajectory. A public bench detects your phone's Wi-Fi signature. A transit gate logs your travel pattern. None of this is required for lighting, seating, or mobility. These are surplus operations, used to extract value. The city becomes a factory that converts everyday life into monetizable information.

Hal Varian, Google's chief economist, articulated the economic strategy. He observed that computer-mediated transactions allow every interaction to be monitored, recorded, and analyzed¹². The smart city reframes the entire urban realm as computer-mediated. Walking a

¹² Zuboff, S. (2019). *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power*. PublicAffairs.

public street becomes an involuntary commercial transaction. The sensor-sidewalk functions like a webpage tracking clicks. The citizen becomes a source of exchange value without ever giving consent. This dynamic marks the emergence of “Technofeudalism”, Yanis Varoufakis’ term for a new economic model, where citizens pay a digital rent of behavioral surplus in exchange for basic urban mobility and access¹³. The infrastructure by itself demands participation in a data market.

Shoshana Zuboff calls this “instrumentarian power”—a form of dominance that aims to shape human behavior through automated tracking and feedback loops¹². Unlike other forms of power, instrumentarian power operates with “radical indifference”¹². The surveillance apparatus does not care about the citizen’s thoughts, emotions, or values. It cares about the quantity of harvestable signals. By tracking every pause, stride, and glance, the system converts human experience into behavioral data, which is then again used to predict and steer the human.

The weaponization of sensing reshapes the relationship between individual and city. The promise of transparency *for* the citizen becomes transparency *of* the citizen. The sensory layer of the feedback loop that was meant to make urban systems smarter is feeding a planetary-scale apparatus of behavioral prediction. Human experience becomes a monetized asset.

2.2 Prediction

Collected data by itself does not yet do much. To fuel the engine of global surveillance, prediction, and control, it must pass through the next stage: reasoning. Algorithmic reasoning instrumentalizes the harvested data. The data profiles stream into vast cloud computation infrastructures: datacenters. They process raw behavioral signals with a serious goal: to predict human behavior. Reasoning is not just analysis. The machine does not try to understand. It tries to anticipate and automate. Reasoning is a statistical attack on future autonomy.

¹³ Varoufakis, Y. (2023). *Technofeudalism: What Killed Capitalism*. Melville House.

Raw data points go through a primary process of classification. These systems do not only sort behaviors into existing categories. They actively produce the categories themselves. Kate Crawford describes this process as an act of world-making¹⁴. The creator of classifications defines what is visible, normal, deviant. This power is centralized in those who label training data and set metrics. A decision about which patterns matter shapes what the city sees and acts upon. Classification is never innocent. It is a political act that hardens assumed realities into computational truths.

The cloud apparatus enabling this reasoning concentrates immense computing power in a handful of corporate firms. Amazon, Google, Microsoft, and a few others own the planetary stacks that process urban life¹⁵. Their infrastructure turns local behavioral surplus into prediction products on a global scale. The reasoning of your street corner is a centralized function, operated remotely, serving commercial and state interests far removed from the public it reshapes.

The underlying ambition goes one step further than knowing the future: it is to act on the future. Predictions are instruments designed to intervene before it happens. The system evaluates the citizen as a behavioral probability cloud. It takes statistical bets on your next movement, your next purchase, your next social contact, and potentially your likelihood to protest. It does not wait for confirmation. It decides based on a projected version of you and acts on that phantom. Forecasts trigger real-world consequences. The system reroutes traffic to manage a predicted jam that has not yet occurred. It denies a loan because a model anticipates future default. It act before you get the chance to do so yourself.

The smart city narrows the gap between prediction and action, reducing the moments of conscious choice. The city becomes the co-author of your decisions. You are a target for remote control. The goal is certainty, the elimination of the unpredicted. Human choice becomes a

¹⁴ Crawford, K. (2021). *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*. Yale University Press.

¹⁵ Bratton, B. H. (2015). *The Stack: On Software and Sovereignty*. MIT Press.

friction to be removed. Shoshana Zuboff identifies the core imperative: to automate us¹⁶. Citizens are not invited to reflect. They are manipulated. The smart city's reasoning functions as a choreographer, scripting urban life ahead of time. Its choreographic directions are passed forward, transforming statistical expectation into physical control.

2.3 Control

Predictions find their application in the execution of preemptive control. Automatic bollards retract for authorized vehicles and snap up against those the system flags as anomalous. Acoustic devices flood plazas with high-frequency tones that only younger ears can hear, forcing away youth before they come together. At transit hubs, facial-recognition turnstiles autonomously sort permitted from excluded. Automated traffic systems control intersections and traffic lanes to prevent a predicted disturbance. The feedback loop completes itself through preemptively triggered actuators. Statistical expectations translate directly into kinetic force.

The prediction, born in the data center, bleeds into the city. It is the physical manifestation of what Gilles Deleuze defined as a society of control, where physical enclosures are replaced by continuous network modulations that fragment the citizen into a dividual—a digitized profile subject to constant tracking and coded gates¹⁷.

Rem Koolhaas identified an ideology behind this shift. He argued that the smart city trades the democratic trinity of liberty, equality, fraternity for a technocratic trinity: comfort, security, sustainability¹⁸. Comfort, here, means no friction—a city scrubbed of surprise. Security leads to predictive policing, the neutralization of predicted crime before it happens. Sustainability, the most seductive, is used as a code word for efficiency: more renewable energy rather than less total power, faster global processing rather than shorter transport distances,

¹⁶ Zuboff, S. (2019). *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power*. PublicAffairs.

¹⁷ Deleuze, G. (1992). Postscript on the societies of control. *October*, 59, 3–7.

¹⁸ Koolhaas, R. (2014, September 24). *Smart cities: The future of urbanism?* [Lecture]. Lecture on Smart Cities, Brussels, Belgium.

additional sensor infrastructure rather than dismantling resource-intensive systems. It is lastly an institutional mandate to optimize growth rather than reduce consumption. Absent from comfort, security, sustainability is any leftover of democratic debate. Decisions and actions become a computational problem, not a political one.

The cost is high. Liberty disappears when a system that claims to "know" your intent stops you before you can act. Equality fails as glass doors and turnstiles sort the permitted from the excluded. Fraternity suffers when public space is turned into a logistical corridor optimized for throughput—no loitering, no friction, no "other." The technocratic city of comfort, security, and sustainability is an ambient, architectural machine that reshapes what is possible to think, feel, and do in public. It might run perfectly, but it runs without its citizens.

Adam Greenfield calls this the frictionless trap¹⁹. By erasing every physical obstacle, every unplanned encounter, every moment of productive resistance, the smart city sterilizes the public realm. What remains is a regime of "conditioned compliance". Hans Teerds deepens the critique: genuine public space is precisely the place where one meets "the other"—the stranger, the dissident, the body that disrupts your expectations²⁰. When hostile design and predictive actuation preemptively smooth that terrain, they kill the encounter required for politics to emerge. What is lost is not merely spontaneity but the very practice of negotiating difference—negotiation that itself becomes a threatening bug in the system's code. In its place stands a given contract: accept the algorithm's choreography, and you may pass without incident.

The full loop does something even more radical than enforcing a predicted, predetermined order. It manufactures the reality it claims to discover. The predictive city becomes a self-fulfilling mechanism. Sensing harvests behavior; reasoning forecasts a future; actuation makes the forecast become real. The system realizes that its prediction came true and with reinforced confidence continues *constructing* reality. When a motorized bench retracts at a predicted time, people will not sit. When a traffic system reroutes vehicles away from a predicted

¹⁹ Greenfield, A. (2017). *Radical Technologies: The Design of Everyday Life*. Verso Books.

²⁰ Teerds, H. (2020). It's a smart world? A critical stance on the smart city. *Architectural Review*, 147(4), 12–19.

jam, the jam will not happen. When the police arrest a predicted suspect before a crime, this person will not commit a crime.

These examples are illustrative but they demonstrate a simple calculation. By preemting the future systems manifest their predictions and therefore reinforce themselves. Predictions become reality and the system can never evaluate what would have happened without manipulation. The possibility of a wrong prediction is erased, and with it, natural urban life: friction, chance, spontaneity, serendipity. In doing so, the loop projects the data from the past into the future to influence the present. It constantly reproduces the past, making it increasingly difficult to move forward. It effectively predicts the future at the cost of tomorrow.

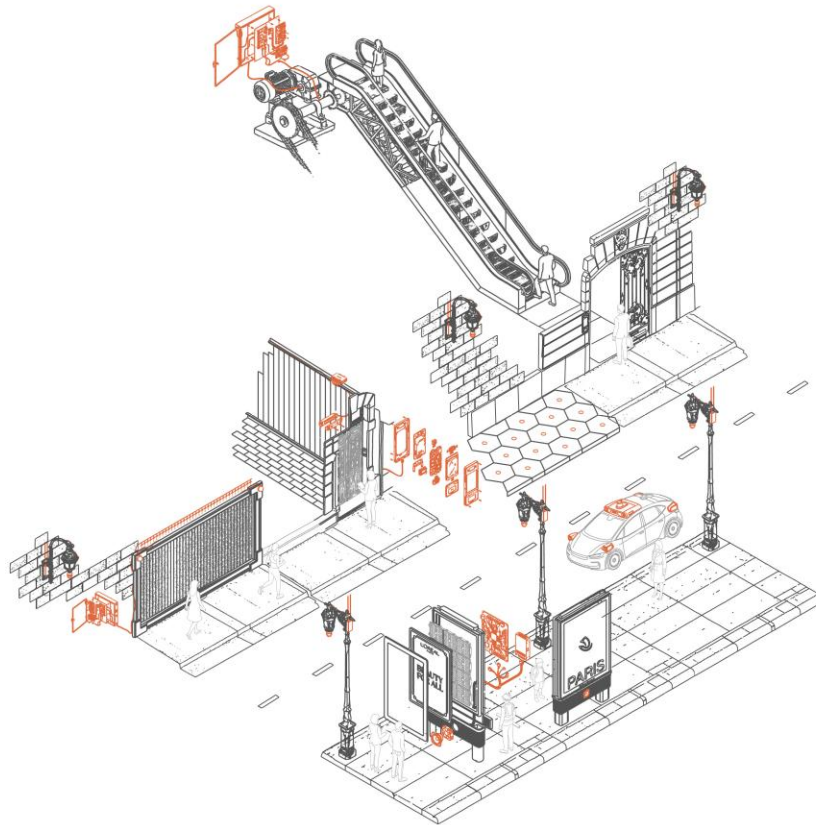


Image 9: The intrusion of “smart” devices in the urban fabric

Chapter 3 : Reversing the Cyborg Premise

Manfred Clynes and Nathan Kline introduced the term “cyborg” in their 1960 article “Cyborgs and Space”²¹. They defined it as “a man-machine system in which the control mechanisms of the human portion are modified externally by drugs or regulatory devices so that the being can live in an environment different from the normal one.” The concept was born from space-travel research. Its logic was soon adapted to life on Earth. Two principles defined this early cyborg. First, technology served as an extension designed to enhance the human body. Second, the biological and the mechanical existed in tight codependence. Donna Haraway later adopted the cyborg as an emblem of boundary transgression²². In her reading, the figure dissolved the binary between organism and machine, nature and artifice. It claimed a radical freedom of self-definition.

Today, the cyborg acquires a new meaning. It is no longer a futuristic dream. It has become a reality with the potential to protect us. The danger lies in the loss of codependence. While deductive algorithms dominated the deeply cyborgian reality of the early twenty-first century, the shift toward inductive machine learning has created a new dispositif. Artificially intelligent cities no longer extend human capability. They operate as autonomous systems, independent of the human. Citizens are reduced to a mine from which behavioral data is extracted.

Gilles Deleuze foresaw this shift. In his Postscript on the Societies of Control he argued that individuals are replaced by “dividuals”²³—subjects divided into digital pieces - data profiles. This division is happening at scale. The smart city operates as a localized node of a

²¹ Clynes, M. E., & Kline, N. S. (1960). Cyborgs and space. *Astronautics*, 5(9), 26–27, 74–76.

²² Haraway, D. (1991). *Simians, Cyborgs, and Women: The Reinvention of Nature*. Routledge.

²³ Deleuze, G. (1992). Postscript on the societies of control. **October**, 59, 3–7.

planetary computational structure. Citizens become what Yanis Varoufakis terms “cloud serfs”²⁴. They generate behavioral data by walking through the street. Movements, habits, and social interactions are converted into cloud capital. This capital is then used to refine the algorithms that guide their behavior—a feedback loop that eliminates the possibility of spontaneous public life.

This situation demands a reversal of the cyborg premise. Instead of attaching digital prosthetics to the human body, the proposal augments computational infrastructure with the biological human. The citizen refuses the role of a behavioral resource. Instead, people become the irrational element that disrupts the system. They push dreams, desires, unreason, and quirks into the machinery that tries to sterilize them. Restored as protagonists of the urban narrative, they contribute the indeterminate and the unpredictable. The reversed cyborg injects human weirdness as a glitch. It reclaims the city by overloading the system with the very unreason it used to filter out. It forces the computational apparatus to deal with the whole spectrum of human complexity.

Three principles guide this reversal: participation, transparency, and friction. Participation emancipates the act of being sensed. It transforms data extraction into a conscious, voluntary physical performance. Transparency opens the black box, exposing algorithmic reasoning to collective oversight. Friction reintroduces deliberate resistance into public space and nurtures confrontation with the other. Together, these principles reform predictive systems. They shift from tools of compliance into platforms for spatial negotiation.

3.1 Participation

Hannah Arendt distinguished between labor, work, and action. Behavior, the term used by technological systems, belongs to the realm of predictable, statistically manageable data. It is exactly what the predictive city extracts and operationalizes. Action, by contrast, resists this condition. It is the human capacity to interrupt automatic processes and to insert the

²⁴ Varoufakis, Y. (2023). **Technofeudalism***: What Killed Capitalism*. Melville House.

unexpected²⁵. Conscious participation replaces passive behavior with action. When sensing becomes a deliberate performance, it transforms the act of being read into an act of writing back.

Participation thrives in a system that has no means of filtering inputs. A person gives whatever they choose to give: a laugh, a sob, a tremor. Dreams, intuitions, and nonsense become fertilizer for algorithmic irrationality. The human body turns into a source of doubt. It forces the computational infrastructure to deal with the biological. Compelled to rationalize this intentional “noise,” the predictive engine hallucinates correlations. It leaves the peak of the standardization curve. It wanders through the system’s own dreams, born from the human body’s refusal to comply.

3.2 Transparency

Transparency must be radical. It unlocks the black box that tech companies would never willingly open. Algorithmic reasoning derives its power precisely from remaining hidden. Without sight, there is no knowledge, no understanding, no resistance. Opacity protects what Shoshana Zuboff calls “instrumentarian power”—the capacity to shape behavior through hidden means of predictability²⁶.

Radical transparency dismantles not just the secrecy of code but the reduction that machine learning enforces. As Kate Crawford argues, the act of classification is a violent reduction of the world to a narrow set of training labels. She calls this process epistemological flattening. It is a political “world-making” operation that erases context and history²⁷. Radical transparency exposes these classifications. It allows citizens to contest the categories that govern their lives.

²⁵ Arendt, H. (1958). **The Human Condition**. University of Chicago Press.

²⁶ Zuboff, S. (2019). **The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power**. PublicAffairs.

²⁷ Crawford, K. (2021). **Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence**. Yale University Press.

The operations of a system that governs everyday environments must become visible to everyone. The early World Wide Web distributed information openly and at incredible speed. In the same spirit, data extracted from public space and from all people must remain available in real time. It should never become a privatized asset. Data ceases to be an extractive artifact. It becomes a public record for negotiation. Reasoning turns into a visible, collective act, collapsing the distance between algorithmic calculation and civic life.

3.3 Friction

The corporate smart city promotes a fantasy of seamlessness. It erases friction as an error. Adam Greenfield calls this the “frictionless trap”—a regime where conflict is eliminated as inefficiency and all social tension is smoothed away²⁸. But the public space of a city is not a platform to be smoothed out. It must resist optimization for speed or throughput. The very idea of optimizing public space is counterproductive. A commitment to democratic life demands public spaces that remain free, open to appropriation, and emancipatory.

Freedom, appropriation, and emancipation produce friction. This friction is not a defect. Political and social friction are essential for public life. Hans Teerds frames friction as the spatial requirement for encountering “the other”—the stranger, the dissident²⁹. Just as a muscle strengthens under load, civic sensitivity grows when physical and social resistance challenge it.

François Roche designed an architecture to actively cultivate this resistance. His concept of “transitory uncertainty”³⁰ embraces the incomplete, the conflicted, and the unmeasurable. Roche’s structures refuse fixed, optimized layouts. The built environment remains indeterminate, forcing inhabitants to constantly negotiate their physical presence.

²⁸ Greenfield, A. (2017). **Radical Technologies: The Design of Everyday Life**. Verso Books.

²⁹ Teerds, H. (2021). **It’s a smart world? An Architectural Reading of ‘Toronto Tomorrow’ through Hannah Arendt’s Notion of the World**. Conference paper presented at the Arendt Circle.

³⁰ Roche, F. (2005). **I’ve heard about: Group of adaptive, transitory scenarios in which the operational mode is uncertainty**. Manifesto.

This spatial negotiation is the foundation for political debate. Maarten Hajer warns that corporate smart urbanism depoliticizes the city³¹. It hides complex social struggles behind the facade of green optimization and comfortable efficiency. On the contrary, Hajer calls for a “democratic theater” where spatial friction stages public debate. Friction, then, is the fuel of civic life - the very matter of spatial negotiation.

³¹ Hajer, M., & Dassen, T. (2014). **Smart About Cities: Visualizing the Challenge for 21st Century Urbanism**. NAI010 Publishers.

Conclusion: Reconfiguring the Feedback Loop

The smart city is an engineered feedback loop. It senses, it reasons, it acts. Under prevailing economic logic, this loop is weaponized. Sensing serves surveillance, the hidden extraction of behavioral data. Reasoning serves prediction, the probabilistic calculation of a constructed future. Action serves control, the automation of the present based on this constructed future. Human complexity is flattened into calculable metrics. Citizens are reduced to passive data sources. Public space becomes a smooth logistical channel with optimized throughput.

This dissertation does not propose escape from the infrastructure itself. It proposes a reversal of the intention. The same feedback loop is repurposed. Sensing becomes a conscious, voluntary act. It demands participation as intentional performance to invite human action. Reasoning is made public and transparent. The algorithmic black box opens to collective examination. Decisions become visible, contestable, a shared space of negotiation. Action is deliberately infused with friction. Spatial interventions provoke the unexpected, resisting optimization and automation. The built environment intensifies as a stage for political encounter.

These three conditions reverse the cyborg premise. Beyond augmenting the human with technology, they augment technology with the human. Human, here, does not mean a passive, reduced, calculable metric of a predictive system. Human, on the other hand, means a complex being that is far greater than the sum of its data profiles. Human, much further, means an organism that includes mistakes, peculiarities, unreasons, and desires. Augmenting smart city technology with such humans means injecting "Voluntary Unreason" directly into the feedback loop. The machinery that forces humans to adapt to its seamless rhythms is being forced to adapt to human complexity.

Architecture is the physical medium that makes this reversal possible. It is the guiding substrate for smart infrastructure. It is the spatial dispositiv that structures and protects spaces of appearance and negotiation. The design of physical environments stages the instruments of participation, the interfaces of transparency, and the mechanisms of friction. Architecture has the power to defend the irrational, the unpredictable, and the collective against algorithmic control. In doing so, it secures the city as a fundamentally human domain—contested, situated, and irreducibly alive.

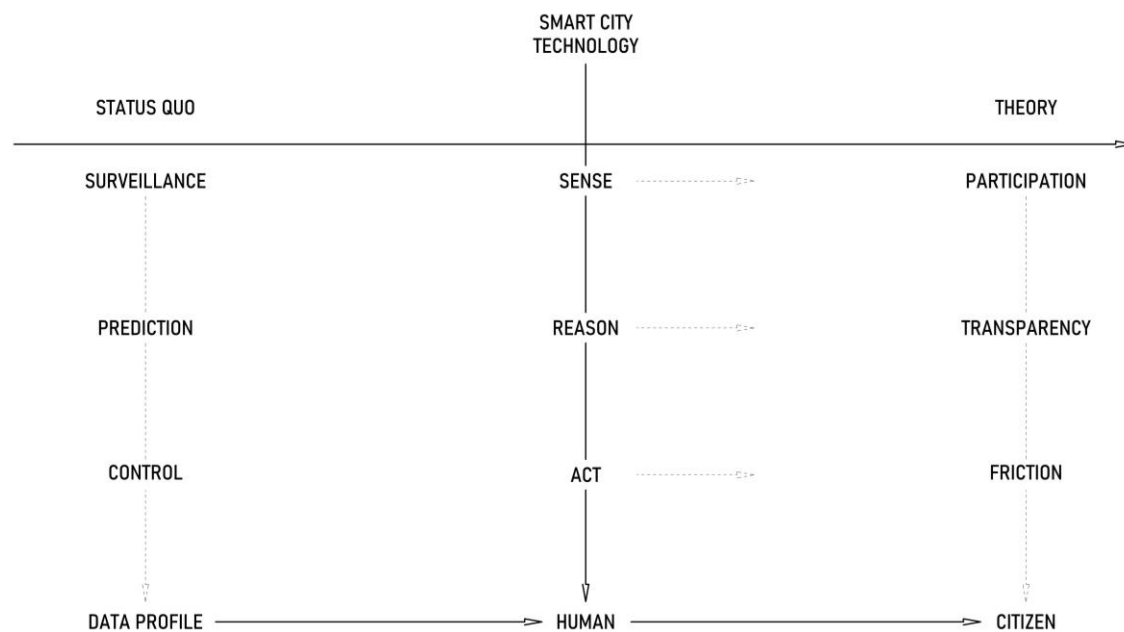


Image 00: Diagram to organize the key chapters of the argumentation into a grid of intentionality.

Glossary

Actuator A physical mechanism that converts digital computational signals into kinetic force or material adjustments. Actuators alter the physical environment—through motors, valves, or light. They close the cyber-physical feedback loop.

Algorithm A coded, sequential set of instructions designed to perform a specific task. Predictive algorithms have shifted from deductive rules to inductive machine learning, calculating probabilities based on data.

Algorithms: Deductive Algorithm Computational systems operating on explicit, pre-programmed, top-down rules (e.g., "if X, then Y"). Historically dominant, these systems execute deterministic commands.

Algorithms: Inductive Algorithm Machine-learning systems that analyze massive datasets to identify historical patterns and calculate future statistical probabilities. They construct their own operational logic, predicting behavior before it occurs.

Artificially Intelligent Cities Urban environments integrated with sensing, reasoning, and acting systems that autonomously predict and regulate human behavior. They shift the city from a static shelter to an active, cybernetic system.

Behavioral Surplus A concept (coined by Shoshana Zuboff) describing behavioral data harvested from human experience that exceeds what is required for service improvements. This surplus is treated as a free raw material, turned into marketable prediction products, and traded in speculative behavioral futures markets.

Cloud Serfdom An economic and spatial condition (articulated by Yanis Varoufakis) where smart-city citizens produce behavioral surplus merely by existing inside the city's spaces. Their lived experience is harvested as a digital rent-payment in exchange for basic urban mobility and access.

Collective Negotiation The social and political process of democratic interaction, debate, and conflict resolution through which the collective of many individuals shares and governs a public space. It is the fundamentally human alternative to the automated decision-making of algorithmic systems.

Computer-Mediated Transaction An economic framework (articulated by Hal Varian) where digital infrastructures monitor, record, and analyze interactions. In platform urbanism, the public street is recoded as a computer-mediated interface, turning the physical act of walking into an involuntary commercial transaction.

Control The capacity to manage, direct, or actuate a system; in cybernetics, the final stage of a feedback loop where calculated decisions are executed to alter an environment. Control is typically disguised as comfort or security.

Cyber-Physical Systems (CPS) Engineered networks that seamlessly integrate digital computational algorithms with physical components, creating continuous interaction, making them inseparable.

Cyborg A cybernetic organism; a biological entity augmented with technological or mechanical prosthetics to enhance or extend its natural capabilities.

Data Profile A quantified, algorithmic representation of an individual. Created from the continuous extraction and aggregation of observable behavioral data. They are the base for probability calculations.

Dividual The algorithmic fragmentation of the unified human subject into digitized data profiles (coined by Gilles Deleuze). In predictive networks, the citizen is no longer an individual, but a divided collection of behavioral signals subjected to continuous modulation.

Epistemological Flattening An algorithmic reductionism (concept by Kate Crawford) where the complex, contextual, and messy realities of human life and physical space are isolated from their context and flattened into standardized numerical data signals, rendering them computable for predictive systems.

Friction The resistance, inefficiency, or unpredictability inherent in physical, social, or mechanical interactions; the fundamental antithesis of seamless, automated processes. Friction is inherent in genuine human interaction. It leads to interaction with “the other”, resists seamless algorithmic control, and sustains democratic public life.

Frictionless Trap A spatial condition (coined by Adam Greenfield) where urban designers erase all physical obstacles, unexpected encounters, and moments of resistance from public spaces. This produces a seamless experience, a sterile city, a landscape of conditioned compliance that suppresses active citizenship.

Instrumentarian Power A theoretical concept (coined by Shoshana Zuboff) describing a species of power that engineers human behavior toward external market ends through ubiquitous, data-driven digital architectures, operating with "radical indifference" to human meaning.

Interface Any point of contact or shared boundary between two complex systems that governs the conditions of exchange and communication between them (Benjamin Bratton).

Parasitism A systemic relationship in which one entity (the parasite) attaches to and derives nutrition from a host infrastructure, often fundamentally altering the host's behavior or resources.

Participation The active, voluntary, and conscious involvement of individuals in a shared process, environment, or political system.

Polar Inertia A theoretical concept (coined by Paul Virilio) describing a condition of profound physical immobility resulting from the instantaneous, global transmission of information, tele-presence, and electromagnetic speed.

Prediction The use of experience, historical data, statistics, and algorithms to forecast the probability of future outcomes or behaviors.

Predictive Architecture Built environments that integrate spatial design with predictive technologies, allowing the physical space to autonomously anticipate and adapt to user behavior before an event occurs.

Predictive Technology Digital systems that harvest vast data sets and apply machine learning algorithms to anticipate, model, and preempt future events or actions.

Preemption (Preemptive Governance) An active logic that intervenes to govern a predicted future. This closes the gap between statistical probability and physical actuation. The system acts on a projected version of the citizen, removing the moment of conscious choice.

Quantified The state of having been measured, counted, and reduced to numerical values, enabling complex phenomena to be computationally processed.

Reversed Cyborg A conceptual inversion of the traditional cyborg: a condition wherein a vast technological or mechanical infrastructure is augmented by the integration of unpredictable biological or human characteristics.

Scaffold A structure used to support, shape, or facilitate the construction and operation of another, typically larger or more important, system.

Societies of Control A paradigm (articulated by Gilles Deleuze) where continuous network modulations replace physical disciplinary enclosures. Concrete barriers yield to biometric tracking and digital code-gates, turning the open city into an invisible, dynamic enclosure.

Space of Appearance A political-philosophical concept (coined by Hannah Arendt and adapted by Hans Teerds) describing the public realm that materializes where bodies gather to act and speak in common. In the smart city, it emerges not through frictionless optimization, but through physical resistance, metabolic effort, and vocal performativity.

Surveillance The continuous, often hidden, observation and monitoring of individuals, groups, or environments, typically for the purpose of data extraction or behavioral management.

Technocratic Trinity An ideological framework (identified by Rem Koolhaas) where smart-city developers trade the democratic values of *liberty, equality, and fraternity* for the operational metrics of *comfort, security, and sustainability*. Decisions are no longer political questions of collective negotiation but computational problems.

Technofeudalism An economic regime (articulated by Yanis Varoufakis) where digital platform monopolies overthrow traditional capitalism. Profits are replaced by digital rent, reducing smart-city citizens to cloud serfs who trade behavioral surplus for spatial access.

The Stack A planetary-scale computational grid (formulated by Benjamin Bratton) comprising six vertical layers: Earth, Cloud, City, Address, Interface, and User. This vertical sovereign apparatus replaces horizontal national organization, restructuring physical and political space.

Transgression The deliberate crossing of established physical, political, or technological boundaries to claim agency and self-definition.

Transitory Uncertainty A design methodology (inspired by François Roche) that prioritizes absolute indeterminacy and adaptive spatial scenarios over automated optimization. Built biostructures continuously shift, contract, or expand, forcing citizens into active physical negotiation with their environment.

Transparency The quality of being open, legible, and visible; in algorithmic systems, the degree to which internal operations, decision-making processes, and data usage are made comprehensible to external observers.

Unreason The domain of human dreams, desires, and irrational quirks that cannot be standardized or predicted by algorithmic systems.

Voluntary Unreason The intentional performance of human quirks, dreams, and nonsense within predictive systems. By refusing predictable behavior, citizens force computational networks to process biological noise, disrupting algorithmic optimization.

World-Making The active production of reality through data classification (concept by Kate Crawford). Rather than neutrally sorting pre-existing behaviors, algorithmic classifiers decide what is visible, normal, or deviant, thereby hardening political and cultural assumptions into mathematical truths.

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